



清华大学

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习题解析

《计算机网络原理》教学团队

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Wireshark实验2——ARP

- ARP协议在以太网帧头中载荷类型的编号是?
- ARP分组头中，以太网硬件类型编号和IP协议类型编号分别是?
- ARP请求分组中，操作码 (Opcode) 值是? 源IP地址及MAC地址，目的IP地址及MAC地址是多少?
- ARP回复分组中，操作码 (Opcode) 值是? 源IP地址及MAC地址，目的IP地址及MAC地址是多少?

```
> Frame 22: 56 bytes on wire (448 bits), 56 bytes captured (448 bit
- Ethernet II, Src: NewH3CTechno_37:88:02 (94:29:2f:37:88:02), Dst:
  > Destination: Intel_85:b2:45 (bc:38:98:85:b2:45)
  > Source: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  Type: ARP (0x0806)
  [Stream index: 0]
  Trailer: 8801006beb19c86bd9238d0950e8
- Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  Sender IP address: 183.173.136.1
  Target MAC address: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Target IP address: 183.173.142.204
```

Arp Reply

```
> Frame 21: 42 bytes on wire (336 bits), 42 bytes captured (336
- Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: Br
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Type: ARP (0x0806)
  [Stream index: 1]
- Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Sender IP address: 183.173.142.204
  Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
  Target IP address: 183.173.136.1
```

Arp Request



Wireshark实验2——思考题

- ping 同一局域网内的主机和局域网外的主机，都会产生 ARP 报文么？所产生的 ARP 报文有何不同，为什么？
 - 都会产生。访问局域网内的是对应问对应主机arp，访问局域网外的是问网关的arp。
- ARP 请求数据包是支撑 TCP/IP 协议正常运作的广播包。如果滥发或错发 ARP 广播包会产生那些不良影响？如何发现和应对？
 - 滥发可能造成DoS等问题，错发可能造成流量劫持。滥发可以检测arp包的发送频率，滤除异常的arp包，错发可以通过arp一致性检查，静态arp绑定等方法处理



Wireshark实验3——IPv4

- Version字段的值是多少？ IHL字段的值一般是多少？ 结合 IPv4 分组头部格式可以看出 IHL 的单位是什么？
 - 4, 5, 单位是4bytes
- 三个分组的 Identification 值是多少？ 是否相等？
 - 图中是56552, 相等
- 前两个分组的Flag字段, DF和MF的值分别是？ 表示什么意思？
 - DF是0, 表示允许分片。MF是1, 表示还有后续分片

```
> Frame 6: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{...}
> Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
Type: IPv4 (0x0800)
[Stream index: 0]
> Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0xdce8 (56552)
> 001. .... = Flags: 0x1, More fragments
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 183.173.142.204
Destination Address: 182.61.200.110
[Reassembled IPv4 in frame: 8]

> Frame 7: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{...}
> Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
Type: IPv4 (0x0800)
[Stream index: 0]
> Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0xdce8 (56552)
> 001. .... = Flags: 0x1, More fragments
...0 0000 1011 1001 = Fragment Offset: 1480
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 183.173.142.204
Destination Address: 182.61.200.110
[Reassembled IPv4 in frame: 8]

> Frame 8: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{29A043BC...}
> Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
> Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
Type: IPv4 (0x0800)
[Stream index: 0]
> Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 68
Identification: 0xdce8 (56552)
> 000. .... = Flags: 0x0
...0 0001 0111 0010 = Fragment Offset: 2960
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 183.173.142.204
Destination Address: 182.61.200.110
> [3 IPv4 Fragments (3008 bytes): #6(1480), #7(1480), #8(48)]
```



Wireshark实验3——IPv4

- 第三个分组的Flag字段，DF和MF的值分别是？表示什么意思？
 - DF是0，表示允许分片。MF是0，表示没有后续分片
- 三个分组Fragment offset的值依次为多少，以确保在乱序到达时也能正确重组出原来的分组？
 - 0, 1480, 2960
- 三个分段的总的数据长度为1500+1500+68-3*20=3008，比ping命令中的参数 3000 多了 8，为什么？
 - 多了8字节ICMP Header

```
> Frame 6: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{...}
Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Type: IPv4 (0x0800)
  [Stream index: 0]
  > Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1500
    Identification: 0xdce8 (56552)
    > 001. .... = Flags: 0x1, More fragments
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 183.173.142.204
    Destination Address: 182.61.200.110
    [Reassembled IPv4 in frame: 8]

> Frame 7: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{...}
Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Type: IPv4 (0x0800)
  [Stream index: 0]
  > Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1500
    Identification: 0xdce8 (56552)
    > 001. .... = Flags: 0x1, More fragments
    ...0 0000 1011 1001 = Fragment Offset: 1480
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 183.173.142.204
    Destination Address: 182.61.200.110
    [Reassembled IPv4 in frame: 8]

> Frame 8: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{29A043BC}
Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Type: IPv4 (0x0800)
  [Stream index: 0]
  > Internet Protocol Version 4, Src: 183.173.142.204, Dst: 182.61.200.110
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 68
    Identification: 0xdce8 (56552)
    > 000. .... = Flags: 0x0
    ...0 0001 0111 0010 = Fragment Offset: 2960
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 183.173.142.204
    Destination Address: 182.61.200.110
    > [3 IPv4 Fragments (3008 bytes): #6(1480), #7(1480), #8(48)]
```



Wireshark实验3——IPv6

- Version字段值是多少，源地址和目标地址分别是什么，占多少字节？
- 观察并说明ipv6是如何进行分片的？
 - 用扩展头进行分片，扩展头包含MF，offset，Next Header等信息
- 观察ipv4与ipv6头部的不同点，比如前文提到的地址长度区别和分段方法区别，除此以外举出1个点辅以截图说明即可。
 - IPv6头部没有checksum

```
> Frame 83: 1510 bytes on wire (12080 bits), 1510 bytes captured (12080 bits) on interface \Device\NPF_{25
Ethernet II, Src: Intel_85:b2:45 (bc:38:98:85:b2:45), Dst: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Destination: NewH3CTechno_37:88:02 (94:29:2f:37:88:02)
  > Source: Intel_85:b2:45 (bc:38:98:85:b2:45)
  Type: IPv6 (0x86dd)
  [Stream index: 0]
Internet Protocol Version 6, Src: 2402:f000:3:8801:107a:40d0:9f83:89fe, Dst: 240c:c0a9:100d::2
  0110 ... = Version: 6
  > .... 0000 0000 .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
  Payload Length: 1456
  Next Header: Fragment Header for IPv6 (44)
  Hop Limit: 128
  > Source Address: 2402:f000:3:8801:107a:40d0:9f83:89fe
  > Destination Address: 240c:c0a9:100d::2
  [Stream index: 6]
Fragment Header for IPv6
  Next header: ICMPv6 (58)
  Reserved octet: 0x00
  0000 0000 0000 0... = Offset: 0 (0 bytes)
  .... .... .... .00. = Reserved bits: 0
  .... .... .... ...1 = More Fragments: Yes
  Identification: 0x39e1ea9c
  [Reassembled IPv6 in frame: 85]
Data (1448 bytes)
  Data [...]: 8000403b000100036162636465666768696a6b6c6d6e6f70717273747576776162636465666768696a6b6c6d6e6f
  [Length: 1448]
```



Wireshark实验3——思考题

- 什么情况下 IPv4 分组需要分段？在哪里分段？又是在哪里重新组装起来的？
 - 当 IPv4 数据报的总长度 > 路径上某条链路的 MTU时，就需要分段，最终在目的主机进行重组。与之对比IPv6只能在源主机分片
- ipv6的头部不包含checksum，ipv6分组是如何进行数据完整性检验的呢？
 - 通过二层或四层的checksum检验



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